Least-squares estimation: from Gauss to Kalman

The Gaussian concept of estimation by least squares, originally stimulated by astronomical studies, has provided the basis for a number of estimation theories and techniques during the ensuing 170 years—probably none as useful in terms of today's requirements as the Kalman filter

H. W. Sorenson *University of California, San Diego*

This discussion is directed to least-squares estimation theory, from its inception by Gauss[1] to its modern form, as developed by Kalman.[2] To aid in furnishing the desired perspective, the contributions and insights provided by Gauss are described and related to developments that have appeared more recently (that is, in the 20th century). In the author's opinion, it is enlightening to consider just how far (or how little) we have advanced since the initial developments and to recognize the truth in the saying that we "stand on the shoulders of giants."

The earliest stimulus for the development of estimation theory was apparently provided by astronomical studies in which planet and comet motion was studied using telescopic measurement data. The motion of these bodies can be completely characterized by six parameters, and the estimation problem that was considered was that of inferring the values of these parameters from the measurement data. To solve this problem concerning the revolution of heavenly bodies, the method of least squares was invented by a "young revolutionary" of his day, Karl Friedrich Gauss. Gauss was 18 years old at the time of his first use of the least-squares method in 1795.

As happens even today (e.g., the Kalman filter), there was considerable controversy in the early 19th century regarding the actual inventor of the least-squares method. The conflict arose because Gauss did not publish his discovery in 1795. Instead, Legendre independently invented the method and published his results in 1806 in his book *Nouvelles méthodes pour la determination des orbites des comètes*. It was not until 1809, in his book *Theoria Motus Corporum Coelestium*, that Gauss published a detailed description of the least-squares method. However, in this treatise Gauss mentions Legendre's discussion of least squares and pointedly refers to his own earlier use (p. 270, *Theoria Motus*)*: "Our principle, which we have made use of since the year 1795, has lately been published by Legendre in the work *Nouvelles méthodes pour la determination des orbites des cometes, Paris, 1806*, where several other properties of this principle have been explained which, for the sake of brevity, we here omit." This reference angered Legendre who, with great indignation, wrote to Gauss and complained[3] that "Gauss, who was already so rich in discoveries, might have had the decency not to appropriate the method of least-squares." It is interesting to note that Gauss, who is now regarded as one of the "giants" of mathematics, felt that he had been eclipsed by Legendre and wrote to a friend saying,[3] "It seems to be my fate to concur in nearly all my theoretical works with Legendre. So it is in the higher arithmetic, . . . , and now again in the method of least-squares which is also used in Legendre's work and indeed right gallantly carried through." Historians have since found sufficient evidence to substantiate Gauss' claim of priority to the least-squares method, so it is Legendre rather than Gauss who was eclipsed in this instance and, indeed, in general.

* The page numbers here refer to the English translation available from Dover Publications, Inc.[1]

The method of least squares

The astronomical studies that prompted the invention of least squares were described by Gauss in *Theoria Motus*.[1] The following quotation (p. 249) not only describes the basic ingredients for Gauss' studies but captures the essential ingredients for all other data-processing studies. "If the astronomical observations and other quantities on which the computation of orbits is based were absolutely correct, the elements also, whether deduced from three or four observations, would be strictly accurate (so far indeed as the motion is supposed to take place exactly according to the laws of Kepler) and, therefore, if other observations were used, they might be confirmed but not corrected. But since all our measurements and observations are nothing more than approximations to the truth, the same must be true of all calculations resting upon them, and the highest aim of all compu-

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tations made concerning concrete phenomena must be to approximate, as nearly as practicable, to the truth. But this can be accomplished in no other way than by a suitable combination of more observations than the number absolutely requisite for the determination of the unknown quantities. This problem can only be properly undertaken when an approximate knowledge of the orbit has been already attained, which is afterwards to be corrected so as to satisfy all the observations in the most accurate manner possible."

Let us briefly reconsider some of the ideas contained in the preceding statement and relate them to "modern" developments.

1. Gauss refers to the number of observations that are absolutely required for the determination of the unknown quantities. The problem of establishing this minimum number of observations is currently discussed in terms of the "observability of the system" and is the subject of many papers; see Refs. 4 and 5, for example.

2. Gauss notes that more observations are required than this minimum because of the errors in the measurements and observations. Thus, he notes the need for "redundant" data to eliminate the influence of measurement errors.

3. Gauss implies that the equations of motion must be exact descriptions, and therefore the problem of dynamic modeling of the system is raised.

4. Gauss requires that approximate knowledge of the orbit be available. This is currently required in virtually all practical applications of Kalman filter theory,[6] for example, and implies the use of some linearization procedure.

5. Gauss states that the parameter estimates must satisfy the observations in the most accurate manner possible. Thus, he calls for the residuals (that is, the difference between the observed values and the values predicted from the estimates) to be as small as possible.

6. Gauss refers to the inaccuracy of the observations and indicates that the errors are unknown or unknowable and thereby sets the stage for probabilistic considerations. In doing so, he anticipates most of the modern-day approaches to estimation problems.

7. Finally, Gauss refers to the "suitable combination" of the observations that will give the most accurate estimates. This is related to the definition of the structure of an estimation procedure (i.e., linear or nonlinear filtering) and to the definition of the performance criterion. These are extremely important considerations in current discussions of estimation problems.

As stated earlier, Gauss invented and used the method of least squares as his estimation technique. Let us consider Gauss' definition of the method (Ref. 1, page 260). He suggested that the most appropriate values for the unknown but desired parameters are the *most probable values*, which he defined in the following manner: ". . . *the most probable value* of the unknown quantities will be that in which the sum of the squares of the differences between the actually observed and the computed values multiplied by numbers that measure the degree of precision is a minimum." The difference between the observed and computed measurement values is generally called the *residual*.

To make the discussion more precise, consider the following statement of the estimation problem. Suppose that m measurement quantities are available at discrete instants of time ($t_1, t_2, \ldots, t_n$) and are denoted at each time t_k as $\mathbf{z}_k$. Suppose that parameters $\mathbf{x}$ are to be determined from the data and are related according to

$$\mathbf{z}_k = H_k\mathbf{x} + \mathbf{v}_k \tag{1}$$

where the $\mathbf{v}_k$ represent the measurement errors that occur at each observation time. As is seen in Eq. (1), the measurement data and the parameters $\mathbf{x}$ are assumed here to be linearly related, thereby making explicit the assumption that Gauss indicated was necessary in the foregoing quotation.

Denote the estimate of $\mathbf{x}$ based on the n data samples $\{\mathbf{z}_1, \mathbf{z}_2, \cdots, \mathbf{z}_n\}$ as $\hat{\mathbf{x}}_n$. Then, the residual associated with the kth measurement is

$$\mathbf{r}_k \triangleq \mathbf{z}_k - H_k\hat{\mathbf{x}}_n \qquad k = 0, 1, \cdots, n \tag{2}$$

The least-squares method is concerned with determining the most probable value of $\mathbf{x}$ (that is, $\hat{\mathbf{x}}_n$). This most probable value is defined as the value that minimizes the sum of the squares of the residuals. Thus, choose $\mathbf{x}$ so that

$$\boxed{L_n = \frac{1}{2}\sum_{k=0}^{n} [\mathbf{z}_k - H_k\mathbf{x}]^T W_k[\mathbf{z}_k - H_k\mathbf{x}]} \tag{3}$$

is minimized. The elements of the matrixes W_k are selected to indicate the degree of confidence that one can place in the individual measurements. As will be explained more fully in the discussion of the Kalman filter, W_k is equivalent to the inverse of the covariance of the measurement noise.

Gauss with his remarkable insight recognized that the simple statement of the least-squares method contains the germ of countless interesting studies. As he says in *Theoria Motus* (Ref. 1, page 269): "The subject we have just treated might give rise to several elegant analytical investigations upon which, however, we will not dwell, that we may not be too much diverted from our object. For the same reason we must reserve for another occasion the explanation of the devices by means of which the numerical calculations can be rendered more expeditious." Judging by the interest in estimation theory over the years, this statement must stand as one of the greatest understatements of all time. In passing, we note that the Kalman filter can be rightfully regarded as an efficient computational solution of the least-squares method.

Gauss did not merely hypothesize the least-squares method; it is interesting to consider his discussion of the problem of obtaining the "most probable" estimate as an introduction to more modern techniques. First, it is significant that he considered the problem from a probabilistic point of view and attempted to define the best estimate as the most probable value of the parameters. He reasoned that errors in the measurements would be independent of each other, so the joint-probability density function of the measurement residuals can be expressed as the product of the individual density functions

$$f(\mathbf{r}_0, \mathbf{r}_1, \cdots, \mathbf{r}_n) = f(\mathbf{r}_0)f(\mathbf{r}_1)\cdots f(\mathbf{r}_n) \tag{4}$$

Next, he argued that the density $f(\mathbf{r}_k)$ would be a normal density

$$f(\mathbf{r}_k) = \frac{\sqrt{\det W}}{(2\pi)^{m/2}} \exp\left[\tfrac{1}{2}\mathbf{r}_k^T W_k \mathbf{r}_k\right] \tag{5}$$

although he recognized that one never obtains errors of infinite magnitude, and thus Eq. (5) is not realistic.

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x otimo é combinação linear das entradas (minimiza o SQR)
Então o erro do valor ótimo (= x - xotimo) é ortogonal às entradas medidas.

However, he rationalized away this difficulty by stating (page 259) that: "The function just found cannot, it is true, express rigorously the probabilities of the errors for since the possible errors are in all cases confined within certain limits, the probability of errors exceeding those limits ought always to be zero while our formula always gives some value. However, this defect, which every analytical function must, from its nature, labor under, is of no importance in practice because the value of our function decreases so rapidly, when $[\mathbf{r}_k^T W_k \mathbf{r}_k]$ has acquired a considerable magnitude, that it can safely be considered as vanishing."

Gauss proceeded by noting that the maximum of the probability density function is determined by maximizing the logarithm of this function. Thus, he anticipated the *maximum likelihood method*, which was introduced by R. A. Fisher[7] in 1912 and has been thoroughly investigated up to the present time. It is interesting that Gauss rejected the maximum likelihood method[8] in favor of minimizing some function of the difference between estimate and observation, and thereby recast the least-squares method independent of probability theory. However, in maximizing the logarithm of the independent and normally distributed residuals, one is led to the least-squares problem defined by Eq. (3).

Kalman filter theory

Let us now leave the early 19th century and enter the 20th century. Consider, briefly, some of the major developments of estimation theory that preceded the introduction of the Kalman filter. As already mentioned, R. A. Fisher introduced the idea of maximum likelihood estimation and this has provided food for thought throughout the subsequent years. Kolmogorov[9] in 1941 and Wiener[10] in 1942 independently developed a linear minimum mean-square estimation technique that received considerable attention and provided the foundation for the subsequent development of Kalman filter theory.

In Wiener–Kolmogorov filter theory, Gauss' inference that linear equations must be available for the solution of the estimation problem is elevated to the status of an explicit assumption. There are, however, many conceptual differences (as one would hope after 140 years) between Gauss' problem and the problem treated by Wiener and Kolmogorov. Not the least of these is the fact that the latter considered the estimation problem when measurements are obtained continuously, as well as the discrete-time problem. To maintain the continuity of the present discussion, attention shall be restricted to the discrete formulation of Wiener–Kolmogorov (and later Kalman) filter theory.

Consider the problem of estimating a signal $\mathbf{s}_n$, possibly time-varying, from measurement data $(\mathbf{z}_0, \mathbf{z}_1, \cdots, \mathbf{z}_n)$, where the $\mathbf{s}_n$ and the $\{\mathbf{z}_i\}$ are related through knowledge of the cross-correlation functions. Assume that the estimate of $\mathbf{s}_n$, say $\hat{\mathbf{s}}_{n/n}$, is to be computed as a linear combination of the $\mathbf{z}_i$:

$$\hat{\mathbf{s}}_{n/n} = \sum_{i=0}^{n} H_{n,i}\mathbf{z}_i \tag{6}$$

The "filter gains" $H_{n,i}$ are to be chosen so that the mean-square error is minimized; that is, choose the $H_{n,i}$ in such a way that

$$M_n = E[(\mathbf{s}_n - \hat{\mathbf{s}}_{n/n})^T(\mathbf{s}_n - \hat{\mathbf{s}}_{n/n})] \tag{7}$$

is minimized. It is well known[11] that a necessary and sufficient condition for $\hat{\mathbf{s}}_{n/n}$ to minimize M_n is that the error in the estimate $(\tilde{\mathbf{s}}_{n/n} \triangleq \mathbf{s}_n - \hat{\mathbf{s}}_{n/n})$ be orthogonal to the measurement data

$$\boxed{E[\tilde{\mathbf{s}}_{n/n}\mathbf{z}_i^T] = 0 \qquad i = 0, 1, \cdots, n} \tag{8}$$

This is the Wiener–Hopf equation, which is frequently written as

$$\boxed{E[\mathbf{s}_n\mathbf{z}_i^T] = \sum_{j=0}^{n} H_{n,j}E[\mathbf{z}_j\mathbf{z}_i^T]} \qquad i = 0, 1, \cdots, n \tag{9}$$

This equation must be solved for the $H_{n,j}$ in order to obtain the gains of the optimal filter. One can rewrite this as a vector-matrix equation whose solution, theoretically speaking, is straightforward. However, the matrix inversion that is required becomes computationally impractical when n is large. Wiener and Kolmogorov assumed an infinite amount of data (that is, the lower limit of the summation is $-\infty$ rather than zero), and assumed the system to be stationary. The resulting equations were solved using spectral factorization.[9,10,12]

The problem formulated and described here is significantly different from Gauss' least-squares problem. First, no assumption is imposed that the signal is constant. Instead, the signal can be different at each n but can be described statistically by the autocorrelation and cross-correlation functions of the signal and measurement data. Second, instead of arguing that the estimate be the most probable, a probabilistic version of the least-squares method is chosen as the performance index.

It has been found that Eq. (9) is solved in a relatively straightforward manner if one introduces a "shaping filter"[13,14] to give a more explicit description of the signal. In particular, suppose that the signal and measurement processes are assumed to have the following structure. The measurements are described by

$$\begin{aligned} \mathbf{z}_i &= \mathbf{s}_i + \mathbf{v}_i \\ &= H_i\mathbf{x}_i + \mathbf{v}_i \end{aligned} \tag{10}$$

where $\mathbf{v}_i$ is a white-noise sequence (that is, $\mathbf{v}_i$ is both mutually independent and independent of $\mathbf{x}_i$). The system state vector $\mathbf{x}_i$ is assumed to be described as a dynamic system having the form

$$\mathbf{x}_{i+1} = \Phi_{i+1,i}\mathbf{x}_i + \mathbf{w}_i \tag{11}$$

where $\mathbf{w}_i$ represents a white-noise sequence. Note that if the noise $\mathbf{w}_i$ is identically zero and if $\Phi_{i+1,i}$ is the identity matrix, then the state is a constant for all i and one has returned basically to the system assumed by Gauss. With the system described by Eqs. (10) and (11), the known statistics for the initial state $\mathbf{x}_0$, and the noise sequences $\{\mathbf{w}_i\}$ and $\{\mathbf{v}_i\}$, one can proceed to the solution of Eq. (9).

Although the weighting function for the filter can be determined, a new solution must be generated for each n. It seems intuitively reasonable that estimates of $\mathbf{s}_{n+1}$ (or $\mathbf{x}_{n+1}$) could be derived, given a new measurement $\mathbf{z}_{n+1}$, from $\hat{\mathbf{s}}_{n/n}$ and $\mathbf{z}_{n+1}$ rather than from $\mathbf{z}_0, \mathbf{z}_1, \cdots, \mathbf{z}_n, \mathbf{z}_{n+1}$, since $\hat{\mathbf{s}}_{n/n}$ is based on the data $(\mathbf{z}_0, \mathbf{z}_1, \cdots, \mathbf{z}_n)$. In 1955 J. W. Follin[15] at Johns Hopkins University suggested a recursive approach based on this idea, which he carried out for a specific system. This approach had immediate appeal and essentially laid the foundation for the developments that are now referred to as the Kalman filter. It is clear (for example, see Ref. 16, p. 129) that Follin's

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work provided a direct stimulus for the work of Richard Bucy, which led to his subsequent collaboration with Kalman in the continuous-time version of the filter equations.[17] As frequently happens, the time was ripe for this approach, because several other people independently investigated recursive filter and prediction methods; see, for example, Refs. 18 and 19. Also, the method of stochastic approximation[20] was introduced and being studied for related problems[21] during this period.

Kalman published his first paper on discrete-time, recursive mean-square filtering in 1960.[2] It is interesting to note that, analogous to the Gauss–Legendre squabble concerning priority of the least-squares method, there is a difference of opinion concerning the originator of the Kalman filter. Peter Swerling published a RAND Corporation memorandum in 1958[18] describing a recursive procedure for orbit determination. Of further interest is the fact that orbit determination problems provided the stimulus for both Gauss' work and more modern-day developments. Swerling's method is essentially the same as Kalman's except that the equation used to update the error covariance matrix has a slightly more cumbersome form. After Kalman had published his paper and it had attained considerable fame, Swerling[22] wrote a letter to the *AIAA Journal* claiming priority for the Kalman filter equations. It appears, however, that his plea has fallen on deaf ears.

The developments beginning with Wiener's work and culminating with Kalman's reflect fundamentally the changes that have occurred in control systems theory during this period. In the "classical control theory," the emphasis was on the analysis and synthesis of systems in terms of their input–output characteristics. The basic tools used for these problems were the Laplace and Fourier transforms. The original formulation and solution of the Wiener–Kolmogorov filtering problem is consistent with this basic approach. More recent developments have stressed the "state-space" approach, in which one deals with the basic system that gives rise to the observed output. It represents in many ways a return to Gauss' approach, since he referred to the dynamic modeling problem as noted earlier. Also, the state-space approach makes use of difference and differential equations rather than the integral equations of the classical approach. Although the two approaches are mathematically equivalent, it seems to be more satisfying to work with differential equations (probably since dynamical systems are generally described in this manner).

At this point, let us summarize the Kalman filtering problem and its solution. The system that is considered is composed of two essential ingredients. First, the state is assumed to be described by

$$\mathbf{x}_{k+1} = \Phi_{k+1,k}\,\mathbf{x}_k + \mathbf{w}_k \tag{11'}$$

and the measurement data are related to the state by

$$\mathbf{z}_k = H_k\mathbf{x}_k + \mathbf{v}_k \tag{10'}$$

where $\{\mathbf{w}_k\}$ and $\{\mathbf{v}_k\}$ represent independent white-noise sequences. The initial state $\mathbf{x}_0$ has a mean value $\hat{\mathbf{x}}_{0/-1}$ and covariance matrix $P_{0/-1}$ and is independent of the plant and measurement noise sequences. The noise sequences have zero mean and second-order statistics described by

$$\boxed{\begin{aligned} E[\mathbf{v}_k\mathbf{v}_j^T] &= R_k\delta_{kj} \qquad E[\mathbf{w}_k\mathbf{w}_j^T] = Q_k\delta_{kj} \\ E[\mathbf{v}_k\mathbf{w}_j^T] &= 0 \qquad \text{for all } k, j \end{aligned}}$$

where δ_{kj} is the Kronecker delta.

An estimate $\hat{\mathbf{x}}_{k/k}$ of the state $\mathbf{x}_k$ is to be computed from the data $\mathbf{z}_0, \mathbf{z}_1, \cdots, \mathbf{z}_k$ so as to minimize the mean-square error in the estimate. The estimate that accomplishes this is to be computed as an explicit function only of the measurement $\mathbf{z}_k$ and the previous best estimate $\hat{\mathbf{x}}_{k-1/k-1}$. This approach leads to a recursive solution that provides an estimate that is equivalent to the estimate obtained by processing all of the data simultaneously but reduces the data-handling requirements. The estimate of the signal

$$\mathbf{s}_k = H_k\mathbf{x}_k \tag{12}$$

is given by

$$\hat{\mathbf{s}}_{k/k} = H_k\hat{\mathbf{x}}_{k/k} \tag{13}$$

The solution of this recursive, linear, mean-square estimation problem can be determined from the orthogonality principle given in Eq. (8), as well as in a large variety of other ways, and is presented below. This system of equations has come to be known as the Kalman filter. The estimate is given as the linear combination of the estimate predicted in the absence of new data, or

$$\boxed{\hat{\mathbf{x}}_{k/k-1} = \Phi_{k,k-1}\hat{\mathbf{x}}_{k-1/k-1}}$$

and the residual $\mathbf{r}_k$. Thus, the mean-square estimate is

$$\hat{\mathbf{x}}_{k/k} = \Phi_{k,k-1}\hat{\mathbf{x}}_{k-1/k-1} + K_k[\mathbf{z}_k - H_k\Phi_{k,k-1}\hat{\mathbf{x}}_{k-1/k-1}] \tag{14}$$

The gain matrix K_k can be considered as being chosen to minimize $E[(\hat{\mathbf{x}}_k - \hat{\mathbf{x}}_{k/k})^T(\mathbf{x}_k - \mathbf{x}_{k/k})]$ and is given by

$$\boxed{K_k = P_{k/k-1}H_k^T(H_kP_{k/k-1}H_k^T + R_k)^{-1}} \tag{15}$$

The matrix $P_{k/k-1}$ is the covariance of the error in the predicted estimate and is given by

$$\boxed{\begin{aligned} P_{k/k-1} &= E[(\mathbf{x}_k - \hat{\mathbf{x}}_{k/k-1})(\mathbf{x}_k - \hat{\mathbf{x}}_{k/k-1})^T] && (16) \\ &= \Phi_{k/k-1}P_{k-1/k-1}\Phi_{k,k-1}^T + Q_{k-1} && (17) \end{aligned}}$$

The $P_{k/k}$ is the covariance of the error in the estimate $\hat{\mathbf{x}}_{k/k}$.

$$\boxed{\begin{aligned} P_{k/k} &= E[\mathbf{x}_k - \hat{\mathbf{x}}_{k/k})(\mathbf{x}_k - \hat{\mathbf{x}}_{k/k})^T] && (18) \\ &= P_{k/k-1} - K_kH_kP_{k/k-1} && (19) \end{aligned}}$$

Equations (14), (15), (17), and (19) form the system of equations comprising the Kalman filter.[2,6]

Kalman filter theory—a perspective

Let us relate elements of this problem to Gauss' earlier arguments. First, Kalman assumes that the noise is independent from one sampling time to the next. But it is clear from Eq. (5) that this is equivalent to assuming that the residual $(\mathbf{z}_k - H_k\mathbf{x}_k)$ is independent between sampling times and therefore agrees with Gauss' assumption. Next, the noise and initial state are essentially assumed by Kalman to be Gaussian. The linearity of the system causes the state and measurements to be Gaussian at each sampling time. Thus the residual is Gaussian, as Gauss assumed. Therefore, one sees that the basic assumptions of Gauss and Kalman are identical except that the latter allows the state to change from one time to the next. This difference introduces a nontrivial modification to Gauss' problem but one that can be treated within a least-squares framework if the noise $\{\mathbf{w}_k\}$ is considered as the error in the plant model at each stage. In particular, one can formulate the least-squares problem as that of

Desde que sejam ruídos brancos, isso me diz que os valores não tem nenhuma ligação entre si. Ou seja, o erro num instante k não depende do instante no isntante j.

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Kalman considera sistemas que variam seus estados com o tempo. Gauss tratou de um sistema cujos estados são fixos.

choosing the estimates $\hat{\mathbf{x}}_{k/k}$ and the plant errors $\mathbf{w}_k$ to minimize the modified least-squares performance index.

$$L_n = \frac{1}{2}(\mathbf{x}_0 - \mathbf{a})^T M_0^{-1}(\mathbf{x}_0 - a) + \frac{1}{2}\sum_{i=0}^{n}(\mathbf{z}_i - H_i\mathbf{x}_i)^T R_i^{-1}(\mathbf{z}_i - H_i\mathbf{x}_i) + \frac{1}{2}\sum_{i=0}^{n-1}\mathbf{w}_i^T Q_i^{-1}\mathbf{w}_i \quad (20)$$

subject to the constraint

$$\mathbf{x}_k = \Phi_{k,k-1}\mathbf{x}_{k-1} + \mathbf{w}_{k-1} \quad (21)$$

Note that the first term essentially describes the uncertainty in the initial state. If one has no a priori information, then M_0^{-1} is identically zero and the term vanishes. Similarly, if there is no error in the plant equation, Q_i^{-1} is identically zero so this term vanishes. Then Eq. (20) is seen to reduce to Gauss' least-squares problem, as given in Eq. (3). The weighting matrices M_0^{-1}, R_i^{-1}, and Q_i^{-1} represent the matrix inverses of the a priori covariance matrices if a probabilistic interpretation is desired.

One can obtain a recursive solution to the problem of minimizing (20) by noting that

$$L_n = L_{n-1} + \frac{1}{2}(\mathbf{z}_n - H_n\mathbf{x}_n)^T R_n^{-1}(\mathbf{z}_n - H_n\mathbf{x}_n) + \frac{1}{2}\mathbf{w}_{n-1}^T Q_{n-1}^{-1}\mathbf{w}_{n-1} \quad (22)$$

and by then proceeding inductively starting with $n = 0$ to obtain recursion relations for the least-square estimate.[23] If this is done, the Kalman filter equations are obtained. It is then indicated that, for this linear problem, deterministic least-squares estimation theory and the probabilistically based mean-square estimation theory are equivalent. Further, the problem of minimizing Eq. (22) is seen to give the most probable estimate for this system.

Since the Kalman filter represents essentially a recursive solution of Gauss' original least-squares problem, it is reasonable to consider the substance of Kalman's contribution and attempt to put it into perspective. It cannot be denied that there has been a substantial contribution if for no other reason than the large number of theoretical and practical studies that it has initiated. I suggest that the contribution is significant for two basic reasons:

1. The Kalman filter equations provide an extremely convenient procedure for digital computer implementation. One can develop a computer program using the Kalman filter in a direct manner that (initially, at least) requires little understanding of the theory that led to their development. There are well-established numerical procedures for solving differential equations, so the engineer does not have to be worried about this problem. By contrast, the solution of the Wiener–Hopf equation and the implementation of the Wiener–Kolmogorov filter must be regarded as more difficult or there would have been no need for the Kalman filter. Since Gauss was very concerned with the computational aspects of least-squares applications, one can imagine that he would appreciate the computational benefits of the Kalman filter.

2. Kalman posed the problem in a general framework that has had a unifying influence on known results. Further, one can analyze the behavior of the estimates within the general framework and thereby obtain significant insights into the results obtained from computational studies. There has been a veritable "explosion" of theoretical papers that have recognizable roots in Kalman's work and thereby testify to the richness of his formulation.

Finally, a third reason for the popularity might be considered, although it is less tangible in character than the other two. It is worth noting that Kalman recognized the potential of his results, whereas others working in the area either did not or were not as successful in communicating the intrinsic worth of recursive filtering to others. One cannot overemphasize the value of recognizing and successfully communicating significant new results.

The Kalman filter, which assumes linear systems, has found its greatest application to nonlinear systems. It is generally used in these problems by assuming knowledge of an approximate solution (as Gauss proposed) and by describing the deviations from the reference by linear equations. The approximate linear model that is obtained forms the basis for the Kalman filter utilization. Commonly, such applications are accomplished with great success but, on occasion, unsatisfactory results are obtained because a phenomenon known as divergence occurs.[24,25]

Divergence is said to occur when the error covariance matrix P_k computed by the filter becomes unjustifiably small compared with the actual error in the estimate. When P_k becomes small, it causes the gain matrix to become too small and new measurement data are given too little weight. As a result, the plant model becomes more important in determining the estimate than are the data and any errors in the model can build up over a period of time and cause a significant degradation in the accuracy of the estimate. This happens most commonly when the plant is assumed to be error-free (i.e., $\mathbf{w}_k \equiv 0$ for all k). If the model were perfect and contained no random or model errors, then the vanishing of the error covariance matrix would be desirable and would represent the fact that the state could be determined precisely if sufficient redundant data were processed. However, it is naive at best to assume that any physical system can be modeled precisely, so it is necessary to account for model errors. Thus, it has become good practice[6] always to include the plant error or noise term $\mathbf{w}_k$. It should be emphasized that divergence does not occur because of any fault of the filter. If the system were actually linear, Kalman[4] showed that the filter equations are stable under very reasonable conditions. Thus, the divergence is a direct consequence of the errors introduced by the linear approximation.

To reduce approximation errors, the so-called "extended Kalman filter" is generally used in practice. In this case the nonlinear system is linearized by employing the best estimates of the state vector as the reference values used at each stage for the linearization. For example, at time t_{k-1}, the estimate $\hat{\mathbf{x}}_{k-1/k-1}$ is used as the reference in obtaining the transition matrix $\Phi_{k,k-1}$. This approximation is utilized in Eq. (17) to obtain the Kalman error covariance $P_{k/k-1}$. The estimate is given by

$$\hat{\mathbf{x}}_{k/k-1} = \mathbf{f}_k(\hat{\mathbf{x}}_{k-1/k-1}) \quad (23)$$

where $\mathbf{f}_k$ is used to denote the nonlinear plant equation.

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In most cases it is obtained as the solution of an ordinary differential equation that describes the plant behavior. The processing of the data obtained at t_k is accomplished in a similar manner. The estimate $\hat{\mathbf{x}}_{k/k-1}$ serves as the reference for the determination of a linear approximation H_k to the nonlinear measurement equation. The matrix H_k is used in Eqs. (15) and (19) to determine the gain and error covariance matrices K_k and $P_{k/k}$. The filtered estimate is then given by

$$\hat{\mathbf{x}}_{k/k} = \hat{\mathbf{x}}_{k/k-1} + K_k[\mathbf{z}_k - \mathbf{h}_k(\hat{\mathbf{x}}_{k/k-1})] \tag{24}$$

where $\mathbf{h}_k$ is used to denote the measurement nonlinearity; that is, the measurement is assumed to be described by

$$\mathbf{z}_k = \mathbf{h}_k(\mathbf{x}_k) + \mathbf{v}_k \tag{25}$$

Through the use of the extended Kalman filter, one can hope to eliminate or reduce divergence. Note, however, that the $P_{k/k-1}$ and $P_{k/k}$ matrices are still linear approximations of the true error covariance matrices. Further, the nonlinear models $\mathbf{f}_k$ and $\mathbf{h}_k$ are themselves approximations of the actual physical system, so modeling errors can still exist. Thus, the extended Kalman filter does not insure the elimination of the divergence problem.

In a practical application one does not know the error in the state estimate, so there are grounds for uneasiness in using this method. Of course, the same type of problem must be considered in any least-squares application and we can return to Gauss for the means of judging the behavior of the filter. Recall that he said that the estimates should satisfy all the observations in the most accurate manner possible. Thus, one is led to further consideration of the residuals as a measure of filter performance. Kailath[26] pointed out recently that the residual sequence $(\mathbf{z}_k - H_k\hat{\mathbf{x}}_{k/k-1})$ is a white-noise sequence. Since the residual can be computed explicitly, it can be examined at each stage to verify that the residual (or innovations) sequence has the appropriate statistical characteristics. A method of controlling the divergence, based on the residuals, has been proposed[27,28] in which the plant and measurement noise covariance matrices Q_k and R_k are chosen in a manner that is appropriate to cause the residuals to have the desired properties. But this method is essentially the same as choosing the least-squares weighting matrices as a reflection of the accuracy of the measurements (or plant). Thus, the least-squares aspect continues to dominate the practical application of the method.

REFERENCES

1. Gauss, K. G., *Theory of Motion of the Heavenly Bodies*. New York: Dover, 1963.

2. Kalman, R. E., "A new approach to linear filtering and prediction problems," *J. Basic Eng.*, vol. 82D, pp. 35–45, Mar. 1960.

3. Bell, E. T., *Men of Mathematics*. New York: Simon and Schuster, 1961.

4. Kalman, R. E., "New methods in Wiener filtering theory," in *Proc. First Symp. of Engineering Application of Random Function Theory and Probability*. New York: Wiley, 1963, pp. 270–388.

5. Sorenson, H. W., "Controllability and observability of linear, stochastic, time-discrete control systems," in *Advances in Control Systems*, Vol. 6, C. T. Leondes, ed. New York: Academic Press, 1968.

6. Sorenson, H. W., "Kalman filtering techniques," in *Advances in Control Systems*, Vol. 3, C. T. Leondes, ed. New York: Academic Press, 1966.

7. Fisher, R. A., "On an absolute criterion for fitting frequency curves," *Messenger of Math.*, vol. 41, p. 155, 1912.

8. Berkson, J., "Estimation by least-squares and by maximum likelihood," *Proc. Third Berkeley Symp. on Mathematical Statistics, and Probability*, Vol. 1. Berkeley: University of California Press, 1956, pp. 1–11.

9. Kolmogorov, A. N., "Interpolation and extrapolation of stationary random sequences," translated by W. Doyle and J. Selin, Rept. RM-3090-PR, RAND Corp., Santa Monica, Calif., 1962.

10. Wiener, N., *The Extrapolation, Interpolation and Smoothing of Stationary Time Series*. New York: Wiley, 1949.

11. Sorenson, H. W., and Stubberud, A. R., "Linear estimation theory," Chapter 1 in *Theory and Application of Kalman Filtering*; *AGARDograph*, no. 139, 1970.

12. Yaglom, A. M., *An Introduction to the Theory of Stationary Random Functions*, translated by R. A. Silverman. Englewood Cliffs, N.J.: Prentice-Hall, 1962.

13. Bode, H. W., and Shannon C. E., "A simplified derivation of linear least-squares smoothing and prediction theory," *Proc. IRE*, vol. 38, pp. 417–425, Apr. 1950.

14. Zadeh, L. A., and Ragazzini, J. R., "An extension of Wiener's theory of prediction," *J. Appl. Phys.*, vol. 21, pp. 645–655, July 1950.

15. Carlton, A. G., and Follin, J. W., "Recent developments in fixed and adaptive filtering," *AGARDograph*, no. 21, 1956.

16. Bucy, R. S., and Joseph, P. D., *Filtering for Stochastic Processes with Applications to Guidance*. New York: Interscience, 1968.

17. Kalman, R. E., and Bucy, R. S., "New results in linear filtering and prediction theory," *J. Basic Eng.*, vol. 83D, pp. 95–108, 1961.

18. Swerling, P., "A proposed stagewise differential correction procedure for satellite tracking and prediction," Rept. P-1292, *RAND Corp.* Santa Monica, Calif., Jan. 1958; also published in *J. Astronaut. Sci.*, vol. 6, 1959.

19. Blum, M., "Recursion formulas for growing memory digital filters," *IRE Trans. Information Theory*, vol. IT-4, pp. 24–30, Mar. 1958.

20. Robbins, H., and Munro, S., "A stochastic approximation method," *Ann. Math. Stat.*, vol. 22, pp. 400–407, 1951.

21. Kiefer, J., and Wolfowitz, J., "Stochastic estimation of the maximum of a regression function," *Ann. Math. Stat.*, vol. 23, pp. 462–466, 1952.

22. Swerling, P., "Comment on 'A statistical optimizing navigation procedure for space flight,'" *AIAA J.*, vol. 1, p. 1968, Aug. 1963.

23. Sorenson, H. W., "Comparison of Kalman, Bayesian and maximum likelihood estimation techniques," Chapter 6 in *Theory and Application of Kalman Filtering*; *AGARDograph*, no. 139, 1970.

24. Schlee, F. H., Standish, C. J., and Toda, N. F., "Divergence in the Kalman filter," *AIAA J.*, vol. 5, pp. 1114–1120, June 1967.

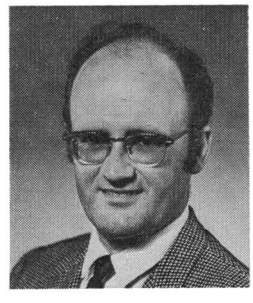
25. Fitzgerald, R. J., "Error divergence in optimal filtering problems," presented at the 2nd IFAC Symposium on Automatic Control in Space, Vienna, Austria, Sept. 1967.

26. Kailath, T., "An innovations approach to least-squares estimation—Part I: linear filtering in additive white noise," *IEEE Trans. Automatic Control*, vol. AC-13, pp. 646–655, Dec. 1968.

27. Jazwinski, A. H., "Adaptive filtering," *Automatica*, vol. 5, pp. 475–485, 1969.

28. Mehra, R. K., "On the identification of variances and adaptive filtering," *Preprints 1969 Joint Automatic Control Conf.*, pp. 494–505.

Harold W. Sorenson (M) was born in Omaha, Nebr., in 1936. He received the B.S. degree in aeronautical engineering from Iowa State University in 1957 and the M.S. and Ph.D. degrees in engineering from the University of California, Los Angeles, in 1963 and 1966, respectively. He worked for General Dynamics/Astronautics from 1957 to 1962 and for the AC Electronics Division, General Motors Corporation, from 1963 to 1966. After a year as guest scientist at the Institut für Luft- und Raumfahrt, Oberpfaffenhofen, West Germany, he joined the faculty of the University of California, San Diego, where he is now an assistant professor of aerospace engineering. His research activities have been centered on linear and nonlinear stochastic control system problems.



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